As the field lines of $\overrightarrow{B_1}$ are passing through the area bounded by solenoid 2, the magnetic flux is linked with each turn of solenoid 2 due to current in solenoid 1 and is given by

$$\Phi_{21} = \int_{A_2} \overrightarrow{B_1}.d\overrightarrow{A} = B_1 A_2 \quad \text{since } \theta = 0^\circ$$
$$= (\mu_\circ n_1 i_1) A_2$$

The flux linkage with solenoid 2 with total turns N_2 is

$$N_2\Phi_{21} = (n_2 l)(\mu_\circ n_1 i_1) A_2 \quad \text{since } N_2 = n_2 l$$
$$N_2\Phi_{21} = (\mu_\circ n_1 n_2 A_2 l) i_1$$

We know that $N_2\Phi_{21} = M_{21} i_1$. Comparing the above equations, we get

$$M_{21} = \mu_\circ n_1 n_2 A_2 l \qquad (4.13)$$

This gives the expression for mutual inductance M_{21} of the solenoid 2 with respect to solenoid 1. Similarly, we can find mutual inductance M_{12} of solenoid 1 with respect to solenoid 2 as given below.

The magnetic field produced by the solenoid 2 when carrying a current i_2 is

$$B_2 = \mu_\circ n_2 i_2$$

This magnetic field B_2 is uniform inside the solenoid 2 but outside the solenoid 2, it is almost zero. Therefore for solenoid 1, the area A_2 is the effective area over which the magnetic field B_2 is present; not area A_1. Then the magnetic flux Φ_{12} linked with each turn of solenoid 1 due to current in solenoid 2 is

$$\Phi_{12} = \int_{A_2} \vec{B}_2 \cdot d\vec{A} = B_2 A_2 = (\mu_\circ n_2 i_2) A_2$$

The flux linkage of solenoid 1 with total turns N_1 is

$$N_1\Phi_{12} = (n_1 l)(\mu_\circ n_2 i_2) A_2 \quad \text{since } N_1 = n_1 l$$
$$N_1\Phi_{12} = (\mu_\circ n_1 n_2 A_2 l) i_2$$

We know that $N_1\Phi_{12} = M_{12} i_2$. Comparing the above equations, we get

$$\therefore M_{12} = \mu_\circ n_1 n_2 A_2 l \qquad (4.14)$$

From equation (4.22) and (4.23), we can write

$$M_{12} = M_{21} = M \qquad (4.15)$$

In general, the mutual inductance between two long co-axial solenoids is given by

$$M = \mu_\circ n_1 n_2 A_2 l \qquad (4.16)$$

If a dielectric medium of relative permeability μ_r is present inside the solenoids, then

$$M = \mu n_1 n_2 A_2 l \text{ (or)}$$
$$M = \mu_\circ \mu_r n_1 n_2 A_2 l$$

EXAMPLE 4.12

The current flowing in the first coil changes from 2 A to 10 A in 0.4 s. Find the mutual inductance between two coils if an emf of 60 mV is induced in the second coil. Also determine the magnitude of induced emf in the second coil if the current in the first coil is changed from 4 A to 16 A in 0.03 s. Consider only the magnitude of induced emf.

Solution

Case (i):

$di_1 = 10 - 2 = 8$ A; $dt = 0.4$ s;

$\varepsilon_2 = 60 \times 10^{-3}$ V

Case (ii):

$di_1 = 16 - 4 = 12$ A; $dt = 0.03$ s

(i) Mutual inductance between the coils.

$$M = \frac{\varepsilon_2}{di_1/dt} = \frac{60 \times 10^{-3} \times 0.4}{8}$$

$$M = 3 \times 10^{-3} \text{ H}$$

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